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Welfare Indicators and Economic Growth in Bangladesh: Evidence from Macro-level Associations and Structural Change

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This study examines the association between economic growth and selected welfare-related macroeconomic indicators in Bangladesh using annual data from 1995 to 2024. Household consumption expenditure growth, employment-to-population ratios, and remittance inflows are considered as indicators reflecting aggregate household economic conditions. The analysis employs descriptive statistics, correlation analysis, Granger causality tests, and a Chow breakpoint test to explore short-run predictive relationships and the stability of welfare–growth linkages over time. The results indicate a strong association and predictive linkage between household consumption growth and GDP per capita growth, while employment and remittance inflows exhibit weaker or statistically insignificant relationships. The Chow breakpoint test identifies a significant structural change around 2020, suggesting instability in established welfare–growth associations during the COVID-19 shock. The findings are exploratory and diagnostic in nature and should be interpreted as indicative macro-level patterns rather than causal effects. Overall, the study provides updated evidence on consumption-growth dynamics and welfare–growth linkages in Bangladesh.

Keywords: Economic Growth; Welfare Indicators; COVID-19; Bangladesh

JEL Classification: O47; I32; E32; O11

1. Introduction

Some of the major challenges faced by emerging economies involve improving household welfare and living standards alongside sustaining economic growth. 8.5 per cent of the world's population is still living on a daily income of merely US$2.15 per day. However, the concern is that efforts to alleviate economic deprivation are stagnating, to the point that 2020–30 is assumed to be a wasted

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decade (World Bank, 2024). At the same time, global economic growth is expected to slow to 2.3 per cent in 2025 (World Bank, 2025a). Although it is often assumed that improvements in living standards accompany economic growth, the impact of development on household welfare improvement remains a matter of debate, as empirical studies often present mixed and contradictory evidence.

For example, according to Ebunoluwa and Yusuf (2018), economic growth is a key instrument for lowering poverty incidence. In contrast, Dollar and Kraay (2002) find no such evidence. Instead, they emphasised that the benefits of growth are almost equal for everyone in the community. On the other hand, Nyasha, Gwenhure, and Odhiambo (2017) discovered that economic growth mitigates poverty-related challenges in the long run only when infant mortality is used as a proxy for household welfare conditions. They also reported evidence of short-term mutual causal connections between economic development and household welfare improvement. Moreover, recent global shocks such as the COVID-19 pandemic have also affected the growth-welfare association. The sufferings caused by the pandemic worsen both economic growth and the poverty scenario (Vitenu-Sackey & Barfi, 2021). While these studies focus primarily on poverty outcomes, the present study adopts a macro-level welfare perspective rather than direct poverty measurement.

However, in Bangladesh, poverty remains a major challenge, despite rapid economic growth. According to The World Bank (2022), 18.7 per cent of Bangladesh's population lived below the national poverty line in 2022. At the same time, 8.01 per cent of people lived below the international poverty line of US$3 per day income, based on the 2021 Purchasing Power Parity. Conversely, Bangladesh experienced growth in real GDP by 6.3 per cent on average yearly, between 2010 and 2023 (World Bank, 2025c). These numbers raise questions about whether economic progress alone is sufficient to improve the living standards of people. Moreover, it is necessary to test for potential structural breaks in the relationship between growth and welfare. This is particularly relevant because external shocks, such as the COVID-19 pandemic, may have altered the dynamics.

Several studies, such as Ferdousi and Dehai (2014), Shahjahan (2014), and Khatun, Shaon, and Sadekin (2021), provide valuable insights into the poverty–growth nexus in Bangladesh. However, most prior studies relied on conventional poverty indicators, such as the headcount ratio, income per capita, and the Gini coefficient. These variables do not fully capture the multidimensional nature of poverty. Moreover, information regarding these variables is not always available.

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Furthermore, very few studies have used remittances as a macro-level welfare indicator in Bangladesh's economy. Moreover, the existing works do not sufficiently analyse the effect of COVID-19 on the growth-welfare relationship in Bangladesh.

Against this background, the present study adopts an alternative perspective by focusing on welfare-related macroeconomic indicators—household consumption expenditure, employment, and remittance inflows—rather than direct poverty measures. These variables reflect household economic conditions and living standards through different channels and are treated as complementary welfare indicators rather than direct proxies for poverty. Using annual data from 1995 to 2024, the study applies descriptive statistics, correlation analysis, Granger causality tests, and a Chow breakpoint test to examine associations, predictive relationships, and potential structural changes around the COVID-19 period.

This paper does not directly measure poverty. Instead, it examines macro-level welfare-related indicators such as household consumption growth, employment-to-population ratios, and remittance inflows. These indicators reflect aggregate household economic conditions and are treated as complementary welfare channels rather than direct poverty measures. Accordingly, the analysis focuses on welfare–growth linkages at the macroeconomic level rather than on causal mechanisms for poverty reduction.

This article contributes to the literature by examining welfare–growth linkages in Bangladesh using a set of welfare-related macroeconomic indicators and updated annual data. Unlike much of the existing work that relies on direct poverty measures or household-level outcomes, the analysis adopts a macro-level welfare perspective and focuses on short-run predictive relationships rather than causal mechanisms. The findings suggest that household consumption growth is more closely associated with GDP per capita growth than with employment or remittance inflows, and that the COVID-19 period coincides with a noticeable structural change in these relationships.

The paper is organised as follows. Section 2 discusses relevant prior studies: Section 3 describes the data and methods used in the analysis; Section 4 interprets the empirical results; Section 5 discusses the findings and their policy relevance; and Section 6 concludes. Therefore, the main objective of this research is to assess the dynamic relationship between economic growth and selected welfare-related macroeconomic indicators in Bangladesh using data from 1995 to 2024. The study

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particularly seeks to examine how welfare-related indicators are associated with economic growth in Bangladesh and whether the COVID-19 period coincides with a structural shift in these relationships.

This paper addresses the following research questions:

1. How are welfare-related indicators—household consumption, employment, and remittances—associated with economic growth in Bangladesh?
2. Do these associations change during periods of major economic disruption, particularly the COVID-19 pandemic?
3. What do these patterns imply for understanding macro-level welfare–growth linkages in a consumption-driven developing economy?

2. Literature Review

2.1 Economic Growth and Household Welfare Outcomes

Economic development is typically regarded as a key component of poverty alleviation; however, its effects vary across countries, contexts, and periods. Škare and Pržiklas Družeta (2016) discovered that poverty cannot be eradicated by growth alone. The rate and pattern of growth, as well as how poverty is measured, all affect how much growth can reduce poverty. According to Hull (2009), whether the poor benefit from growth depends on the structure of the economy and the labour mobility of low-income groups. Similarly, Herman (2011) found that sectoral growth through job creation and labour market flexibility is a key factor in determining how growth translates into welfare. Olaoye (2023) and Donaldson (2008) highlighted the inconsistency of the growth-poverty link, as some economies have experienced rising poverty despite strong growth, while others have reduced poverty with limited growth.

2.2 Beyond Growth: Distribution, Inflation, Finance, and Policy Drivers

Along with GDP growth, several other factors also substantially influence the nexus between growth and household welfare outcomes. For instance, Shahjahan (2014) observed that while growth is associated with welfare outcomes, inflation diminishes real income and offsets gains in household welfare conditions. While Bigsten et al. (2003) and Olaoye (2023) argued that inequality cancels out the benefits of growth. According to them, fairer income distribution is essential for sustainable welfare conditions. In contrast, Dada and Fanowopo (2020)

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emphasised that economic growth supported by strong institutional quality is associated with higher household consumption and lower poverty levels in the long run. While Uddin et al. (2014) and Abdin (2016) found that financial development indirectly contributes to living standards by facilitating economic progress. These findings suggest that growth effectiveness depends not only on its rate but also on complementary policies and the inclusiveness of economic institutions.

2.3 Empirical Evidence from Developing Economies

Recent studies highlight the diversity of the outcomes of the growth-poverty and growth-welfare nexus across regions. For instance, Ebunoluwa and Yusuf (2018) reported a positive link between growth and improvements in household welfare in Nigeria, but noted that unemployment weakens this effect. On the other hand, Oyewale and Musiliu (2015) found that income and export growth contributed to lower poverty in Nigeria, although not significantly. While Henneh (2023) also observed that, in Ghana, GDP per capita growth is most strongly associated with poverty reduction, other factors, such as inflation, unemployment, or overall growth, have weaker or unclear effects.

In Sub-Saharan Africa, Mogess, Eshete, and Alemaw (2023) found that, alongside growth, previous poverty levels significantly influence current poverty levels, highlighting structural persistence. Aiyemo (2024) also found a positive association between growth and poverty reduction in the SSA region, particularly after the mid-2000s.

Sánchez Dávila (2023) reported that approximately 40 per cent of observed improvements in household welfare were attributed to economic growth.

From a regional development perspective, Naufal and Fikriah (2023) showed that growth and poverty in Aceh Province, Indonesia, are reported to be causally related in both directions, with inequality acting as a stabilising factor. Michálek and Výbošťok (2018) discovered that economic growth is associated with reductions in the percentage of people living below the poverty line across all regions of Slovakia, and has a favourable impact on income disparities.

There is also some evidence from Bangladesh regarding this issue. For instance, Ferdousi and Dehai (2014) reported that, due to inflationary pressures, economic growth in Bangladesh between 1991 and 2010 did not sufficiently improve the condition of the poor. Khatun, Shaon, and Sadekin (2021) also noted that despite consistent GDP growth, the welfare gaps could not be closed sufficiently, even with remittances and employment.

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2.4 Synthesis

Altogether, the literature reveals that though economic growth remains an essential element in poverty reduction and broader household welfare outcomes, its success depends on structural inclusiveness, equitable distribution, and resilience to shocks. Recent studies of Vitenu-Sackey and Barfi (2021) and Gupta et al. (2021) emphasise that crises such as COVID-19 have disrupted the traditional growth-welfare relationships. This implies that the re-examination of these dynamics using updated data and context-specific indicators is necessary. The focus of the current study is thus justified, as rapid economic growth, persistent poverty, and structural vulnerability coexist in Bangladesh.

Despite rich evidence, few studies have jointly examined the predictive and structural dynamics between growth and welfare proxies in Bangladesh using recent data. This gap motivates the present analysis.

3. Methodology

This empirical analysis combines descriptive and econometric approaches to investigate the relationship between economic growth and welfare-related macroeconomic indicators in Bangladesh, using time-series data. This study uses annual data from 1995 to 2024 and four key variables. The variables are: GDP per capita growth (annual %) (proxy for economic development), Employment to population ratio, 15+, total (%) (modelled ILO estimate), Households and NPISHs Final consumption expenditure per capita growth (annual %), and personal remittances received (% of GDP). The latter three variables are treated as indicators reflecting household welfare conditions and living standards. The source of the datasets is World Development Indicators (World Bank, 2025b). The variables selected as welfare-related indicators capture different aspects of quality of life. Collectively, they reflect the dimensions of welfare improvement related to income, consumption, and employment. These dimensions are commonly used in empirical literature related to poverty and growth.

For estimation purposes, first, descriptive statistics and graphical analysis are applied to capture the overall trends and patterns of the variables over time. Then the Pearson Correlation test is employed to measure the strength and direction of the connections among these variables. After that, an independent sample t-test is used to compare the mean values between two periods: 1995–2009 and 2010–2024. This test determines if living standard indicators have changed noticeably over

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time. Then, Granger causality tests are used to examine short-run predictive relationships among the variables. These tests are not intended to establish economic or structural causation. Given the macro-level nature of the data and the absence of a formal identification strategy, the results are interpreted as indicating temporal precedence and predictive associations, not causal effects. Before conducting the Causality test, all necessary preliminary tests were conducted to ensure data suitability. The Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) tests ensured the stationarity. A lag length of one is selected based on the VAR lag length criteria. The diagnostic tests confirmed that the model is stable and free from autocorrelation. Detailed outputs of the unit root, lag selection, and diagnostic tests are provided in Appendices *B*–*D*. However, after all the diagnostic and causality tests, the study proceeds with the Chow breakpoint test to measure the structural shift due to COVID-19. The Chow breakpoint test is applied to examine whether the relationship between economic growth and welfare-related indicators remained stable over time. The year 2020 is selected as the breakpoint to capture the onset of the COVID-19 pandemic, which constituted a major exogenous shock affecting household consumption, labour market conditions, and remittance flows in Bangladesh. The purpose of this test is to assess coefficient stability during this period of disruption, rather than to identify causal effects of the pandemic itself. All of these analyses were performed using Microsoft Excel, SPSS, and Eviews. The combination of descriptive, causality, and structural break analyses allows the study to capture both the long-term trends and the short-run dynamic interactions between growth and welfare indicators, as well as any disruptions in their established relationship. While these methods are appropriate for exploring associations, predictive relationships, and structural stability, they do not allow for causal inference.

4. Data Analysis and Findings

4.1 Descriptive Statistics

The descriptive statistics show a broad picture of Bangladesh's key macroeconomic indicators by providing their average values, variations, and patterns. This helps to understand the country's overall economic performance and structural changes during the study period.

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Table 1: Descriptive Statistics Outcome of the Key Variables

	GDPGR	HCEGR	EMP	REM
Mean	4.49	3.34	56.13	6.12
Median	4.81	3.10	56.01	5.32
Maximum	6.97	8.52	59.03	10.59
Minimum	2.24	-1.91	53.65	2.90
Std. Dev.	1.39	2.47	1.28	2.37
Skewness	-0.08	-0.13	0.69	0.47
Kurtosis	1.66	2.59	3.85	1.95
Jarque-bera	2.26	0.29	3.28	2.50
Probability	0.32	0.86	0.19	0.29
Sum	134.67	100.29	1683.97	183.46
Sum sq. dev.	56.09	176.78	47.67	163.14
Observations	30	30	30	30

Source: Author's calculation

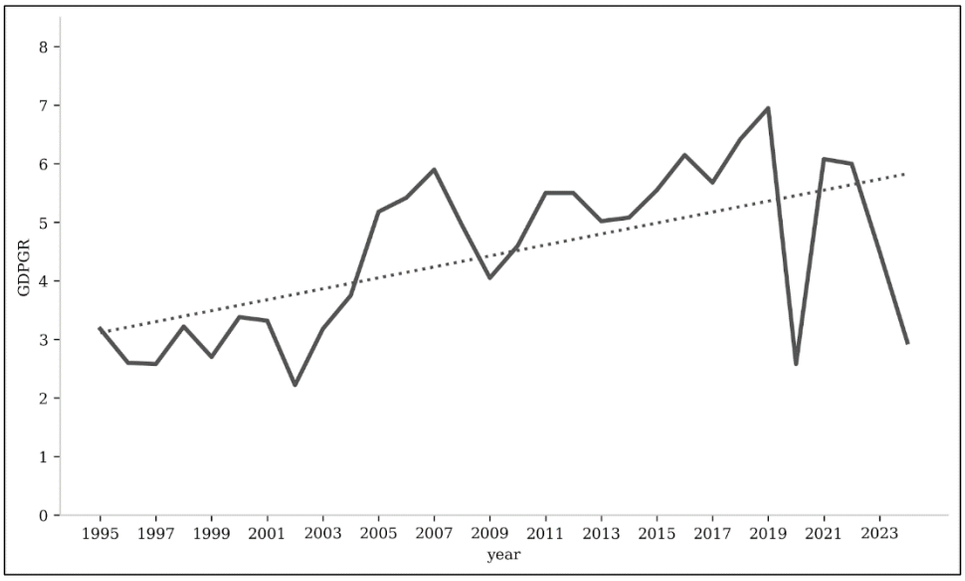
Table 1 presents the descriptive statistics for GDP per capita growth (GDPGR), household consumption expenditure growth (HCEGR), employment-to-population ratio (EMP), and remittances received as a share of GDP (REM) in Bangladesh for the period 1995–2024. The mean value of GDPGR during this period was 4.49 per cent, with values ranging from 2.24 per cent to 6.97 per cent. These numbers indicate consistent yet moderate growth performance. While the average value of HCEGR was 3.34 per cent. It showed relatively higher volatility with the standard deviation of 2.47 and a negative minimum value. Conversely, the average value of EMP was around 56 per cent. This implies that there were limited structural shifts in the labour market during this period. Meanwhile, the average value of REM was about 6.12 per cent of GDP. However, a noticeable fluctuation was also captured by the minimum and maximum values of 2.90 per cent and 10.59 per cent, respectively. Jarque–Bera test results indicate that all variables follow an approximately normal distribution. Braodly, these statistics indicate relatively stable macroeconomic conditions in Bangladesh during the study period, with moderate variation across growth, consumption, employment, and remittance indicators.

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4.2 Trend Analysis

To complement the descriptive findings, trend analysis illustrates how each variable has evolved. This graphical analysis draws attention to long-term trends and movements that support the econometric findings that follow.

Figure 1: Trend of GDP Per Capita Growth (Annual %), 1995-2024

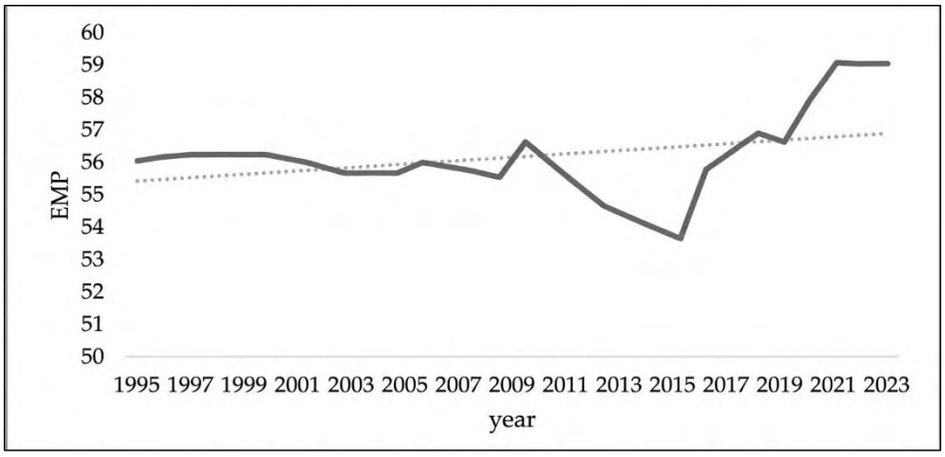


Source: Author's calculation

Figure 1 illustrates the trend of GDP per capita growth (annual %) in Bangladesh over the period 1995–2024. The trend line shows a positive slope. This positively sloped trend line indicates a general upward movement in GDP per capita growth over the study period. However, the figure also captures short-term fluctuations. A sharp decrease around 2020–2021 explains how the COVID-19 pandemic disrupted the economy.

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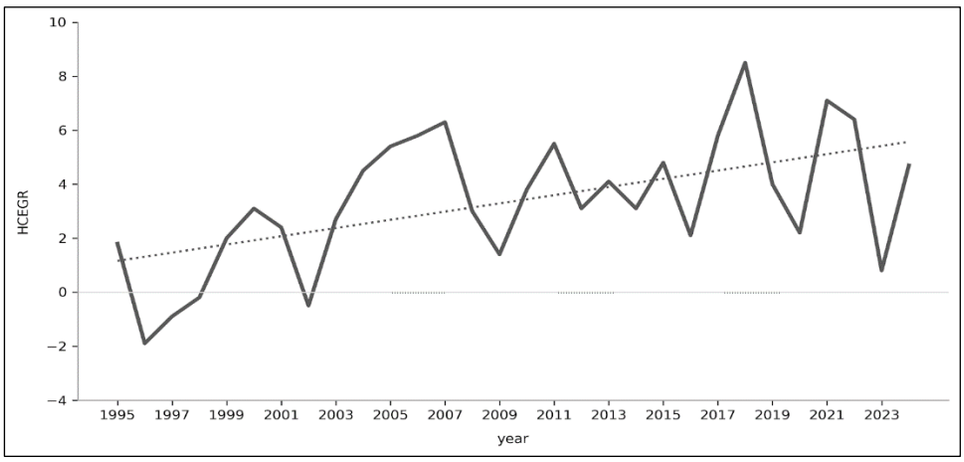
Figure 2: Trend of Employment-to-Population Ratio, 1995-2024



Source: Author's calculation

Figure 2 depicts the trend of the employment-to-population ratio (%) in Bangladesh from 1995 to 2024. The series shows a relatively stable pattern over the years, with only moderate fluctuations around the long-term mean of approximately 56 per cent. A decline between 2010 and 2016 suggests labour market challenges, followed by recovery after 2017. The overall trend indicates a gradual improvement in labour participation.

Figure 3: Trend of Households and NPISHs Final Consumption Expenditure Per Capita Growth, 1995-2024

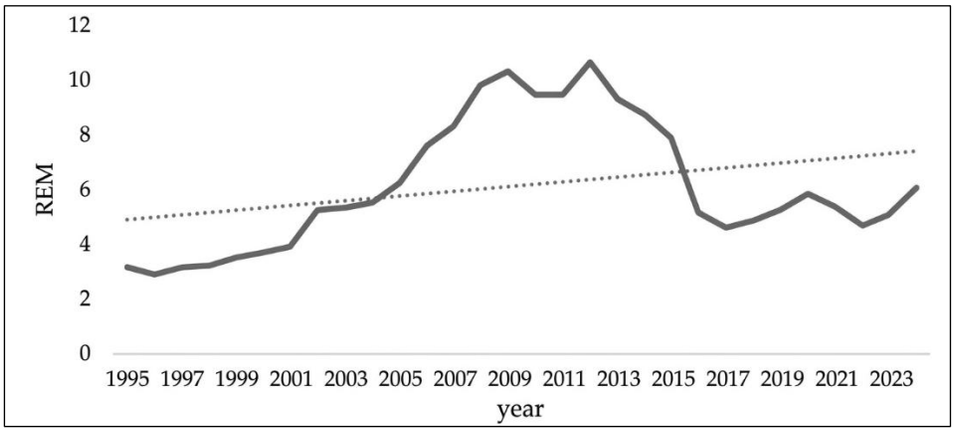


Source: Author's calculation

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Figure 3 presents the trend of household and NPISHs final consumption expenditure per capita growth (annual %) in Bangladesh from 1995 to 2024. The series displays substantial fluctuations. There are periods of strong positive growth, such as 2006–2008 and 2016–2018. At the same time, there are also occasional downturns where consumption growth dropped to the negative territory (e.g., 1996–1997 and 2021–2022). Despite these short-term volatilities, the fitted trend line shows a clear upward trajectory, suggesting gradual welfare improvement alongside rising domestic demand.

Figure 4: Trend of Personal Remittances Received, 1995-2024.



Source: Author's calculation

Figure 4 illustrates the trend of personal remittances as a percentage of GDP in Bangladesh from 1995 to 2024. The graph shows a gradual increase in the remittance inflows until the early 2010s. However, a sharp decline is visible after 2015, followed by a moderate recovery in recent years. Although remittance levels have fluctuated over time, the overall trend remains positive, indicating the continued importance of remittances in the Bangladeshi economy

4.3 Pearson Correlation

While the trend analysis provided insights into the directional movements of GDP, employment, household consumption, and remittances over time, it does not reveal the degree of association between them. Therefore, Pearson correlation is applied to measure the strength and direction of these relationships, as it offers a helpful preliminary step before regression analysis by identifying potential linear linkages among variables.

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Table 2: Pearson Correlation Test Result

	GDPGR	HCEGR	EMP	REM
GDPGR	1	0.7253	-0.1029	0.4140
HCEGR	0.7253	1	0.1165	0.3063
EMP	-0.1029	0.1165	1	-0.3098
REM	0.4140	0.3063	-0.3098	1

Source: Author's calculation

Table 2 reports the Pearson correlation coefficients among the key variables. The results reveal a strong positive correlation between GDPGR and HCEGR, with an r value of 0.73. This indicates that periods of higher economic growth tend to be associated with higher household consumption growth. GDPGR also shows a moderate positive correlation with REM (r = 0.41). In contrast, the relationship between GDPGR and EMP is weak and negative (r = –0.10), suggesting a weak association between output growth and job creation. Employment and remittances are also negatively correlated with the r value of -0.31. Collectively, the results highlight a close association between growth and consumption, while employment and remittances exhibit weaker and more complex relationships.

4.4 Independent Sample t-Test

Correlation analysis indicates the direction and strength of relationships among variables. However, it does not reveal whether these relationships have changed over time. To examine this, an independent samples t-test is applied to compare the two periods and also to find out whether the observed differences reflect statistically significant structural shifts.

Table 3: Findings of the Independent Sample t-Test

Variable	Mean (1995–2009)	Mean (2010–2024)	Mean Diff.	T-value	Sig.(2-tailed)	Significance
GDP growth (%)	3.73	5.25	1.52	–3.56	0.001	***
HCE growth (%)	2.32	4.37	2.05	–2.46	0.020	**
Employment (%)	55.94	56.33	0.39	–0.84	0.410	n.s.
Remittances (% of GDP)	5.44	6.79	1.35	–1.60	0.121	n.s.

Source: Author's calculation
Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; n.s. = not significant.

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Mean differences shown as absolute values (|Group 1 – Group 2|); higher means in the later period (2010–2024) indicate improvement.

The t-test results from Table 3 indicate statistically significant differences in GDP growth and household consumption growth between the two sub-periods. On the other hand, changes in employment and remittance between the two periods are not statistically significant.

4.5 Granger Causality Test

To examine short-run predictive relationships, Granger causality tests were conducted. Unit root tests confirm that all variables become stationary after first differencing, indicating that they are integrated of order one, I (1) (Appendix B). A lag length of one was selected based on VAR criteria, and diagnostic tests confirm model stability (Appendices C and D).

Table 4: Granger Causality Test Outcome

Null Hypothesis	Obs	F-Statistic	Prob.
D_HCEGR does not granger cause D_GDPGR	28	31.1662	8.E-06
D_GDPGR does not granger cause D_HCEGR		3.69104	0.0662
D_EMP does not granger-cause D_GDPGR	28	0.00055	0.9814
D_GDPGR does not granger cause D_EMP		0.58144	0.4529
D_REM does not granger cause D_GDPGR	28	0.12261	0.7292
D_GDPGR does not granger cause D_REM		0.59939	0.4461
D_EMP does not granger cause D_HCEGR	28	0.00388	0.9508
D_HCEGR does not granger cause D_EMP		0.14100	0.7105
D_REM does not granger cause D_HCEGR	28	0.08698	0.7705
D_HCEGR does not granger cause D_REM		0.00189	0.9657
D_REM does not granger cause D_EMP	28	3.15262	0.0880
D_EMP does not granger cause D_REM		0.98181	0.3312

Source: Author's calculation

Table 4 shows a strong unidirectional Granger causality from household consumption growth to GDP growth ($F = 31.17$, $p < 0.01$), indicating that past consumption growth helps predict GDP growth. The reverse relationship is only marginally significant at the 10% level, suggesting weak feedback.

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No significant Granger-causal relationships are observed between GDP growth and either employment or remittances. A weak predictive relationship from remittances to employment is detected at the 10 per cent level. Overall, household consumption growth emerges as the strongest short-run predictor of GDP growth, while employment and remittances play limited roles. These results reflect predictive associations rather than causal effects.

4.6 Chow Breakpoint Test

A Chow breakpoint test was conducted to examine whether the relationship between economic growth and welfare-related indicators remained stable over time. The year 2020 was selected as the breakpoint to capture the impact of the COVID-19 pandemic.

Table 5: Chow Breakpoint Test Result

Test Statistic	Value	Df	Probability
F-statistic	12.12917	(4, 21)	0.0000 ***
Log likelihood ratio	34.71429	–	0.0000 ***
Wald statistic	48.51670	(4)	0.0000 ***

Source: Author's calculation

As shown in Table 5, the test results strongly reject the null hypothesis of stability, indicating a significant structural change in 2020. The OLS estimates suggest that household consumption is significantly associated with GDP growth, while employment and remittances do not show statistically significant associations (see Appendix E). In summary, the findings indicate that the growth–welfare relationship in Bangladesh changed during the COVID-19 period, with household consumption remaining the most closely associated welfare indicator.

5. Discussion

This paper examines how welfare-related indicators—household consumption expenditure, employment, and remittances—are associated with economic growth in Bangladesh during 1995–2024, and whether the COVID-19 pandemic introduced a structural shift in these relationships. The results indicate that household consumption expenditure is closely associated with GDP per capita growth in Bangladesh. Both the regression and Granger causality findings suggest that increases in household consumption growth precede and help predict

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economic growth. Taken together, the results indicate that household consumption growth shows a stronger association with economic growth than other welfare-related indicators. This result is similar to the findings of Chen, Xia, and Wang (2021) and Arouri, Ben Youssef, and Nguyen (2017). Both of them observed that increased household consumption, particularly on essentials such as food, health, and education, is associated with improved household welfare outcomes.

The Granger causality results highlight a unidirectional predictive relationship from household consumption growth to GDP growth, underscoring the importance of domestic demand in sustaining Bangladesh's economic performance. This result is consistent with evidence showing that domestic demand is essential to the economic development of Bangladesh and that policies that increase consumption dynamics are often discussed in relation to economic expansion (Dey & Tareque, 2018). But according to Amin (2011), Economic growth Granger-causes consumption expenditure, meaning that increasing output and incomes result in more household consumption, which is in line with Keynesian consumption theory.

In contrast, employment does not exhibit a significant predictive relationship with GDP growth, suggesting limited short-run transmission from output expansion to job creation. This finding implies that economic growth during the study period may have been less employment-intensive. Similarly, remittances do not show a direct predictive link with GDP growth, indicating that their macroeconomic impact may be limited or indirect. While remittances can improve household welfare at the micro level, they may not have been sufficiently channelled into productive investments at the national level. This is consistent with findings by Eggoh, Bangake, and Semedo (2019), Khan (2024), and Rehman, Shah, and Rehman (2023), who document relatively small growth effects of remittances.

Additionally, the Chow Breakpoint Test points to a structural break in 2020, the year the COVID-19 pandemic began. This structural shift suggests that the pandemic coincided with instability in the established growth–welfare relationship, particularly around household spending and employment. This outcome is consistent with global evidence found in studies. For example, Vitenu-Sackey and Barfi (2021) and Gupta et al. (2021) found that COVID-19 has weakened traditional growth–poverty linkages. However, the sudden drop in

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both growth and welfare indicators highlights how vulnerable the country is to external shocks and the significance of building economic resilience through diversification and social protection.

5.1 Policy Implications

Based on the overall findings, household spending has been more closely associated with Bangladesh's growth. In contrast, employment and remittances show comparatively weaker associations with economic growth and broader welfare stability. This suggests that improving the employment and productivity linkages that drive growth is necessary for sustained household welfare improvement. Policy discussions may consider encouraging the creation of jobs in labour-intensive sectors, increasing opportunities for vocational training, and improving labour market participation, particularly for young people and women. Additionally, investment incentive programs that promote the efficient use of remittance funds can potentially transform immediate consumption advantages into long-term welfare gains. Lastly, strong social safety nets and fiscal strategies that protect household spending are necessary to maintain economic stability during external shocks like COVID-19.

5.2 Limitations

Although the findings provide valuable insights into the growth–welfare linkages in Bangladesh, there are still a number of limitations that should be carefully taken into consideration. First, the study relies on yearly data, which may conceal the micro-level variances in household welfare and short-term fluctuations.

Second, the proxy variables used in the study, such as household consumption, employment, and remittances to represent household welfare improvement, do not capture multidimensional aspects of poverty, such as health, education, or inequality. Third, due to the lack of data availability, other important variables that influence the growth–welfare relationship could not be included.

Future research could address these gaps using higher-frequency data and incorporating additional variables to provide a more comprehensive understanding.

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6. Conclusion

The main purpose of the study is to explore the relationship between economic growth and welfare-related indicators in Bangladesh using annual data over the period 1995–2024. Household consumption, employment, and remittances were used as indicators reflecting household welfare conditions for the investigation. The results indicate that household consumption growth is more closely associated with economic growth than employment or remittance inflows. However, employment and remittance indicators exhibit weaker or statistically insignificant associations with economic growth. The Granger causality results indicate a unidirectional predictive relationship from household consumption growth to GDP growth. This highlights the relevance of domestic demand in understanding Bangladesh's growth dynamics. Finally, the Chow Breakpoint Test identified a significant structural shift in 2020. This structural shift suggests that the COVID-19 pandemic temporarily disrupted the established relationship between growth and welfare indicators. This finding reveals that the economy is sensitive to external shocks and highlights the relevance of policy discussions on building a more resilient economic structure. Taken together, the study suggests that consumption growth has been more closely associated with economic growth than other welfare-related indicators. However, policy approaches such as enhancing employment opportunities and labour productivity, and encouraging productive investment of remittances, may inform discussions on sustainable household welfare outcomes in Bangladesh. By strengthening employment linkages and improving the efficiency of social protection measures, future growth may translate more effectively into broader welfare stability and inclusive development.

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Conflict of Interest

The author declares no conflict of interest

Data Availability Statement

The data that support the findings of this study are obtained from the World Development Indicators (WDI) database maintained by the World Bank https://databank.worldbank.org/source/world-development-indicators. These data are publicly available and properly cited within the manuscript. Processed datasets and analysis code are available from the corresponding author upon reasonable request.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

The author acknowledges the use of tools such as Grammarly, QuillBot, and ChatGPT for writing assistance in preparing the manuscript. Nevertheless, there is no substantial contribution of these tools to the content; thus, these do not qualify for authorship in accordance with the journal guidelines. All ideas, analyses, and conclusions are entirely the author's own, and the author takes full responsibility for the integrity and accuracy of the manuscript.

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Appendix

Appendix A: Data Description and Sources

Variable	Description	WDI Code	Unit	Expected Relationship	Source
GDPGR	GDP per capita growth (annual %)	NY.GDP.PCAP.KD.ZG	%	Proxy for economic development	World Bank (2024)
HCEGR	Household and NPISHs final consumption expenditure per capita growth (annual %)	NE.CON.PRVT.PC.KD.ZG	%	Proxy for living standards	World Bank (2024)
EMP	Employment-to-population ratio, 15+, total (%)	SL.EMP.TOTL.SP.ZS	%	Proxy for labour market participation	World Bank (2024)
REM	Personal remittances received (% of GDP)	BX.TRF.PWKR.DT.GD.ZS	% of GDP	Proxy for external household income	World Bank (2024)

Source: Author's calculation based on World Development Indicators (World Bank, 2024).

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Appendix B: Unit Root Test

	At Level	GDPGR	HCEGR	EMP	REM
UNIT ROOT TEST TABLE (PP)					
With constant	t-Statistic	-2.5302	-3.3445	-0.5393	-1.5825
	Prob.	0.1190	0.0219	0.8692	0.4785
		n0	**	n0	n0
With constant & trend	t-Statistic	-3.0922	-4.1572	-0.8602	-1.3007
	Prob.	0.1268	0.0142	0.9474	0.8676
		n0	**	n0	n0
Without constant & trend	t-Statistic	-0.5219	-1.6225	0.8722	-0.1468
	Prob.	0.4818	0.0976	0.8924	0.6244
		n0	*	n0	n0
	At First Difference	d(GDPGR)	d(HCEGR)	d(EMP)	d(REM)
With constant	t-Statistic	-8.1196	-11.8536	-4.2777	-3.5619
	Prob.	0.0000	0.0000	0.0024	0.0135
		***	***	***	**
With constant & trend	t-Statistic	-11.7182	-18.2731	-4.4786	-3.5587
	Prob.	0.0000	0.0000	0.0070	0.0523
		***	***	***	*
Without constant & trend	t-Statistic	-8.2274	-9.0768	-4.2692	-3.5778
	Prob.	0.0000	0.0000	0.0001	0.0009
		***	***	***	***
UNIT ROOT TEST TABLE (ADF)					
With constant	t-Statistic	-2.6957	-2.2588	-0.3099	-1.8591
	Prob.	0.0869	0.1917	0.9117	0.3457
		*	n0	n0	n0
With constant & trend	t-Statistic	-3.0847	-4.4881	-0.7198	-1.0611
	Prob.	0.1285	0.0069	0.9619	0.9186
		n0	***	n0	n0
Without constant & trend	t-Statistic	-0.7368	-0.2029	0.8722	0.0606
	Prob.	0.3885	0.6038	0.8924	0.6942
		n0	n0	n0	n0

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	At First Difference	d(GDPGR)	d(HCEGR)	d(EMP)	d(REM)
With constant	t-Statistic	-7.0430	-7.6613	-4.2733	-3.5232
	Prob.	0.0000	0.0000	0.0024	0.0148
		***	***	***	**
With constant & trend	t-Statistic	-5.7623	-7.6417	-4.5399	-3.6108
	Prob.	0.0004	0.0000	0.0061	0.0470
		***	***	***	**
Without constant & trend	t-Statistic	-7.1758	-7.6680	-4.2622	-3.5386
	Prob.	0.0000	0.0000	0.0001	0.0010
		***	***	***	***

Note: MacKinnon (1996) provides one-sided p-values for unit root tests. Critical values and probabilities are obtained from MacKinnon (1996).

Appendix C: Lag Selection

VAR Lag Order Selection Criteria
Endogenous variables: D_GDPGR D_HCEGR D_EMP D_REM
Exogenous variables: C
Date: 10/22/25 Time: 20:41
Sample: 1995 2024
Included observations: 27

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-165.6108	NA	3.361517	12.56377	12.75574	12.62085
1	-136.9959	46.63175*	1.343273*	11.62933	12.58920*	11.91475*
2	-119.9618	22.71206	1.359535	11.55273*	13.28051	12.06649

* indicates lag order selected by the criterion

LR: sequential modified LR test statistic (each test at 5% level)

FPE: Final prediction error

AIC: Akaike information criterion

SC: Schwarz information criterion

HQ: Hannan-Quinn information criterion

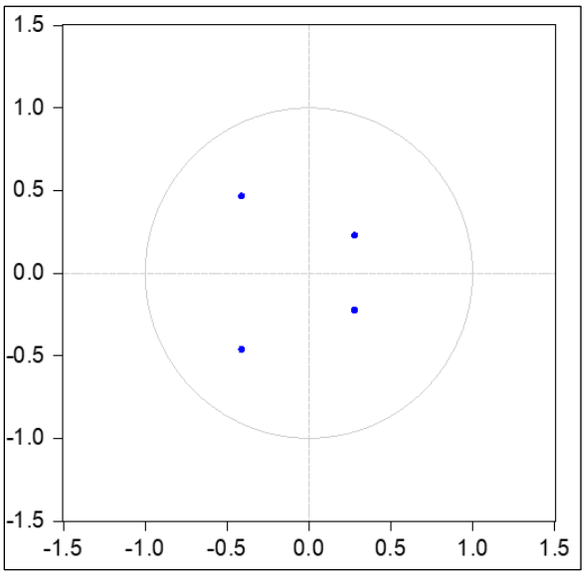
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Appendix D: Diagnostic Tests

Serial Correlation LM:

VAR Residual Serial Correlation LM Tests		
Null Hypothesis: no serial correlation at lag order h		
Date: 10/22/25 Time: 20:54		
Sample: 1995 2024		
Included observations: 28		
Lags	LM-Stat	Prob
1	23.69779	0.0963
Probs from chi-square with 16 df.		

Inverse Roots of AR Characteristic Polynomial



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Appendix E: Chow Break Point Test (with OLS)

OLS:

Dependent Variable: D_GDPGR
Method: Least Squares
Date: 10/22/25 Time: 21:03
Sample (adjusted): 1996 2024
Included observations: 29 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.016213	0.224765	-0.072135	0.9431
D_HCEGR	0.184349	0.088398	2.085441	0.0474
D_EMP	0.195820	0.382201	0.512347	0.6129
D_REM	-0.300960	0.255363	-1.178556	0.2497
R-squared	0.252881	Mean dependent var		-0.007617
Adjusted R-squared	0.163227	S.D. dependent var		1.293171
S.E. of regression	1.182932	Akaike info criterion		3.301311
Sum squared resid	34.98319	Schwarz criterion		3.489903
Log likelihood	-43.86901	Hannan-Quinn criter.		3.360376
F-statistic	2.820631	Durbin-Watson stat		2.684657
Prob(F-statistic)	0.059434			

Chow Breakpoint Test

Chow Breakpoint Test: 2020
Null Hypothesis: No breaks at specified breakpoints
Varying regressors: All equation variables
Equation Sample: 1996 2024

F-statistic	12.12917	Prob. F(4,21)	0.0000
Log likelihood ratio	34.71429	Prob. Chi-Square(4)	0.0000
Wald statistic	48.51670	Prob. Chi-Square(4)	0.0000

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